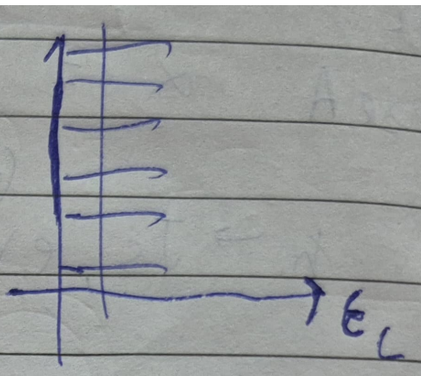
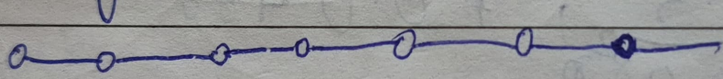
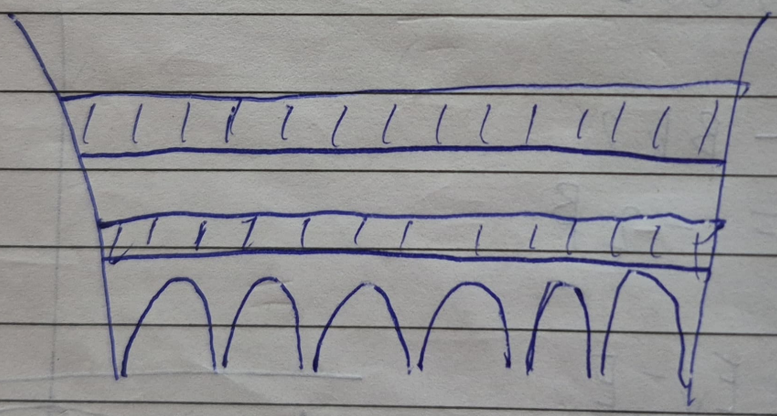
Density of States -



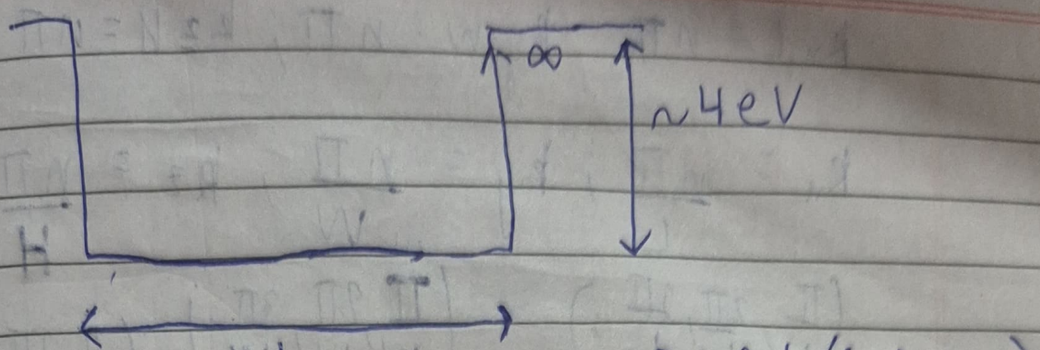
E-K diagram → DOS

or Analytical & closed form expression





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3D solid (L, W, H)

$$-\frac{\hbar^2}{2m}\left[\frac{d^2\psi}{dx^2}+\frac{d^2\psi}{dy^2}+\frac{d^2\psi}{dz^2}\right]=E\psi$$

$$\frac{d^2\psi}{dx^2}+\frac{d^2\psi}{dy^2}+\frac{d^2\psi}{dz^2}+k^2\psi=0 \;;\; k^2=\frac{2mE}{\hbar^2}$$

$$\psi(x,y,z)=\psi_x(x)\,\psi_y(y)\,\psi_z(z)$$

$$\psi_y\psi_z\frac{d^2\psi_x}{dx^2}+\psi_x\psi_z\frac{d^2\psi_y}{dy^2}+\psi_y\psi_x\frac{d^2\psi_z}{dz^2}+k^2\psi_x\psi_y\psi_z=0$$

$$\Rightarrow \frac{1}{\psi_x}\frac{d^2\psi_x}{dx}+\frac{1}{\psi_y}\frac{d^2\psi_y}{dy^2}+\frac{1}{\psi_z}\frac{d^2\psi_z}{dz^2}+k^2=0$$

$$\frac{1}{\psi_x}\frac{d^2\psi_x}{dx}+k_x^2=0 \;;\; \frac{1}{\psi_y}\frac{d^2\psi_y}{dy^2}+k_y^2=0$$

$$\frac{1}{\psi_z}\frac{d^2\psi_z}{dz^2}+k_z^2=0$$

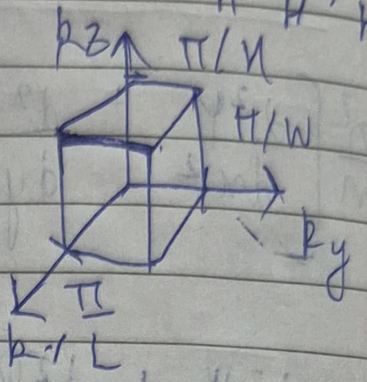
Where $k^2=k_x^2+k_y^2+k_z^2$

$$\Rightarrow \psi_x=A_1\sin k_x x \qquad \psi_z=A_3\sin k_z z$$
$$\psi_y=A_2\sin k_y y$$

$k_x L = n\pi$, $k_y W = n\pi$, $k_z H = n\pi$

$k_x = \frac{n\pi}{L}$; $k_y = \frac{n\pi}{W}$; $k_z = \frac{n\pi}{H}$ (n is integer $\neq 0$)

$(\frac{\pi}{L}, \frac{2\pi}{L}, \frac{3\pi}{L}..)$ $(\frac{\pi}{W}, \frac{2\pi}{W}, \frac{3\pi}{W}..)$ $(\frac{\pi}{H}, \frac{2\pi}{H}, \frac{3\pi}{H}..)$



So $\frac{\pi^3}{LWH}$ contains one state

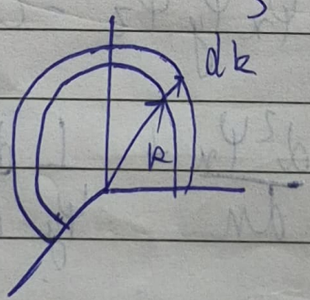
For volume $dk_x dk_y dk_z$:

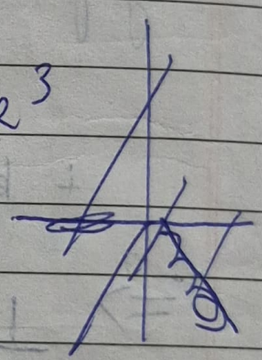
$$\frac{dk_x dk_y dk_z}{(\pi^3/LWH)}$$ no. of states.

Sphere: $\frac{4}{3}\pi(k+dk)^3 - \frac{4}{3}\pi k^3$

$dk_x dk_y dk_z$

$= 4\pi k^2 dk$





No. of states $= \frac{4\pi k^2 dk}{(\pi^3/LWH)}$

Because $n = \neq 1$ has same energy, we consider only 1 octant. (so $\frac{1}{8}$)

Also, $\times 2$ to account for spin (2 different state in same energy state)

$\Rightarrow$ No. of states $= \frac{1}{8} \times \frac{4\pi k^2 dk}{(\pi^3/LWH)} \times 2$ (spin)

remove -ves (duplicates)

No of states $(dN) = \frac{4\pi k^2 dk}{(\pi^3/LWH)} \times \frac{1}{8} \times 2$

Density of states (DOS) $= \frac{1}{V}\left(\frac{dN}{dE}\right)$ $(V = LWH)$

$$DOS = \frac{1}{V}\left(\frac{dN}{dE}\right) = \frac{1}{V}\left(\frac{dN}{dk}\right)\left(\frac{dk}{dE}\right)$$

Assume parabolic dispersion: $E = \frac{\hbar^2 k^2}{2m}$

$$dE = \frac{2\hbar^2 k\, dk}{2m}$$

$$dE = \frac{\hbar^2}{m}\sqrt{\frac{2mE}{\hbar^2}}\, dk$$

$$dN = \frac{4\pi\left(\frac{2mE}{\hbar^2}\right)\frac{m}{\hbar^2}\sqrt{\frac{\hbar^2}{2mE}}\, dE \times \frac{1}{4}}{(\pi^3/V)}$$

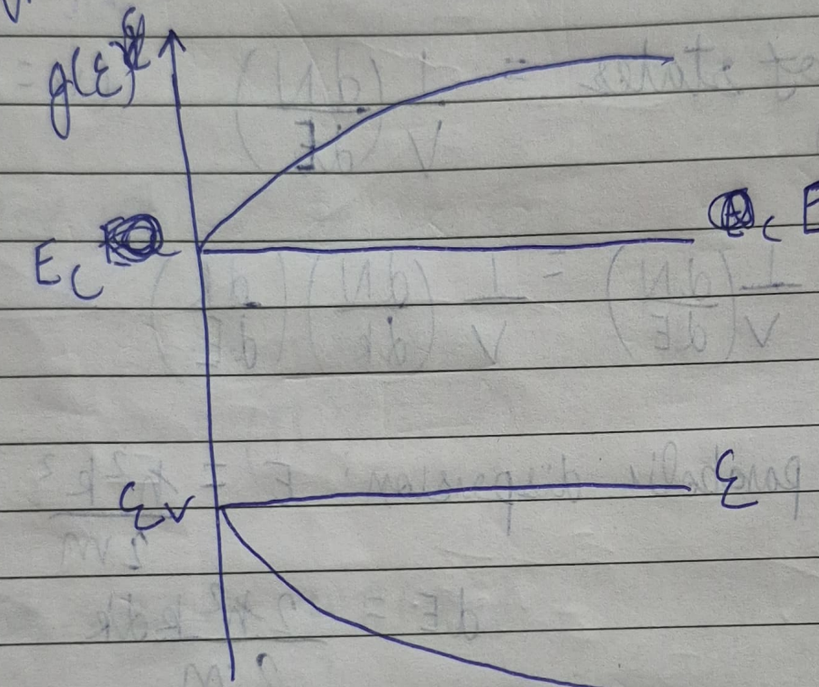
$$\frac{1}{V}\left(\frac{dN}{dE}\right) = \frac{\left(\frac{1}{V}\right)\left(\frac{4\pi}{4}\right)\left(\frac{2mE}{\hbar^2}\right)\left(\frac{m}{\hbar^2}\right)\sqrt{\frac{\hbar^2}{2mE}}}{(\pi^3/V)}$$

$$= \frac{2m^2 E}{\pi^2\hbar^4}\sqrt{\frac{\hbar^2}{2mE}}$$

$$\Rightarrow DOS = \frac{1}{V}\left(\frac{dN}{dE}\right) = \frac{m\sqrt{2mE}}{\pi^2\hbar^3}$$

$$DOS_e = g_n(E) = \frac{m_n^*\sqrt{2m_n^*(E-E_c)}}{\pi^2\hbar^3}$$

$$DOS_h = g_p(E) = \frac{m_p^*\sqrt{2m_p^*(E_v-E)}}{\pi^2\hbar^3}$$



$$N_{(e^-)} = \int_{E_c}^{\text{Top of band}} g_n(E) f(E)\, dE$$

↓ probability

$$p_{(holes)} = \int_{\text{Bottom of valence band}}^{E_v} g_p(E)\,(1-f(E))\, dE$$

Semiconductors in Equilibrium

Equilibrium:

No exchange of energy between system and surroundings

- Intrinsic Semiconductor:

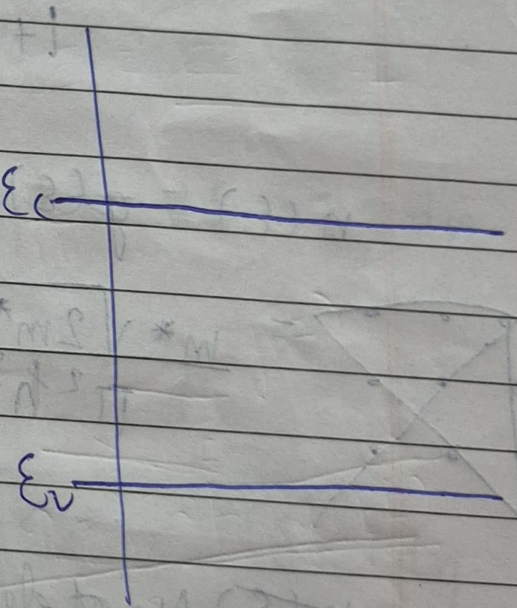
- Highly pure crystalline solid
- Consider a region which is far from surface / interfaces

No. of electron $= \int_0^\infty n(E)\,dE$

$g_c(E)\, f(E)$

$g_c(E)$ → DOS in conduction band

$f(E)$ → Probability of being occupied

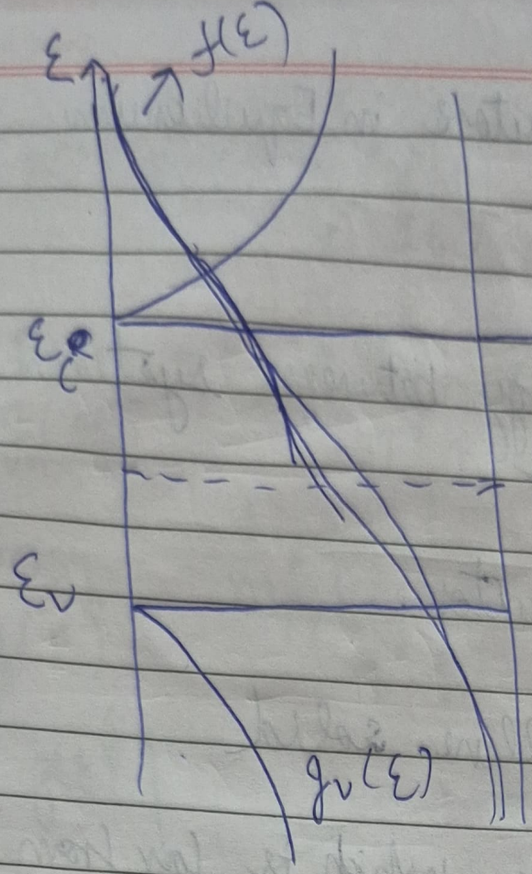


$$g_c(E) = \frac{m^* \sqrt{2m(E-E_c)}}{\pi^2 \hbar^3}$$

m^* → DOS effective mass

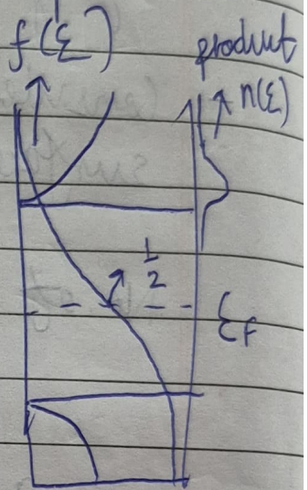
$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{\partial^2 \epsilon}{\partial k_i \partial k_i}$$

↳ Band Structure effective mass



$$f(\varepsilon) = \frac{1}{1+e^{\left(\frac{\varepsilon-\mu}{kT}\right)}}$$

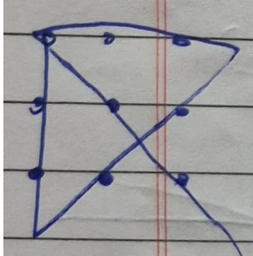
where μ is chemical potential



$\mu = \varepsilon_f$ @ 0K

$$f(\varepsilon) = \frac{1}{1+e^{\left(\frac{\varepsilon-\varepsilon_f}{kT}\right)}}$$

$$n(\varepsilon) = g(\varepsilon) f(\varepsilon)$$



$$= \frac{m^* \sqrt{2m^*(E-E_c)}}{\pi^2 \hbar^3} \cdot \frac{1}{1+e^{\left(\frac{\varepsilon-\varepsilon_f}{kT}\right)}}$$

~~n(E)~~ No. of electrons $= \int_{\varepsilon_c}^{\varepsilon_{c\,top}} \frac{m^* \sqrt{2m^*(E-E_c)}}{\pi^2 \hbar^3} \cdot \frac{1}{1+e^{\left(\frac{\varepsilon-\varepsilon_f}{kT}\right)}} dE$

Maxwell-Boltzmann approximation:

$\varepsilon_c - \varepsilon_f \gg 3kT$

$$\Rightarrow \frac{1}{1+e^{\frac{\varepsilon-\varepsilon_f}{kT}}} \approx \frac{1}{e^{\left(\frac{\varepsilon-\varepsilon_f}{kT}\right)}} = e^{-\left(\frac{\varepsilon-\varepsilon_f}{kT}\right)}$$

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$$n = \int_{E_c}^{E_{c,top}} \frac{m^* \sqrt{2m^*(E-E_c)}}{\pi^2 \hbar^3} e^{-\frac{(\xi-\xi_F)}{kT}} d\xi$$

$\xi_{c,top} - \xi_c|_{Si} = 4eV$; $kT \sim meV$

$\therefore$, assume $E_{c,top} \to \infty$

$$n = \int_{\xi_c}^{\infty} \frac{m^* \sqrt{2m^*(E-E_c)}}{\pi^2 \hbar^3} e^{-\frac{(\xi-\xi_F+E_c-E_c)}{kT}} d\xi$$

$$n = \int_{E_c}^{\infty} \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^*(E-E_c)}\, e^{-\frac{(\xi-\xi_c)}{kT}} e^{-\frac{(\xi_c-\xi_F)}{kT}} dE$$

Let $\frac{E-E_c}{kT} = \eta \to E - E_c = \eta kT$; $dE = kT\, d\eta$

At $E = E_c$; $\eta = 0$; At $E = \infty$, $\eta = \infty$

$$\Rightarrow n = \int_0^{\infty} \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^* \eta kT}\, e^{-\eta} e^{-\frac{(\xi_c-\xi_F)}{kT}} kT\, d\eta$$

$$= m^* \frac{\sqrt{2m^*kT}}{\pi^2 \hbar^3} kT \left(\int_0^{\infty} \sqrt{\eta}\, e^{-\eta} d\eta \right) e^{-\frac{(\xi_c-\xi_F)}{kT}}$$

$\int_0^{\infty} \sqrt{\eta}\, e^{-\eta} d\eta \to \sqrt{\pi}/2$

$$n = \frac{m^* \sqrt{2m^*} (kT)^{3/2}}{\pi^2 \hbar^3} \frac{\sqrt{\pi}}{2} e^{-\frac{(\xi_c-\xi_F)}{kT}}$$

$$n = \frac{\sqrt{2}(m^* kT)^{3/2}}{\pi^2 \left(\frac{h}{2\pi}\right)^3} \frac{\sqrt{\pi}}{2} e^{-\left(\frac{E_c - E_f}{kT}\right)}$$

$$= 4\sqrt{2}\left(\frac{m^* kT\pi}{h^2}\right)^{3/2} e^{-\left(\frac{E_c - E_f}{kT}\right)}$$

$$= 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} e^{-\left(\frac{E_c - E_f}{kT}\right)}$$

↳ N_c (effective density of states)

$$\therefore\ n = N_c \exp\left(-\left(\frac{E_c - E_f}{kT}\right)\right)$$

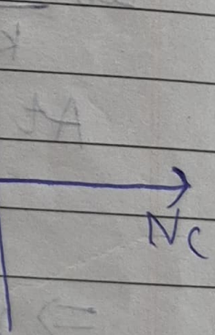
$N_c^{Si} \approx 2.86 \times 10^{19}/cm^3$ (@ 300K)

E_f in this expression is actually μ (chemical potential)

E_f is strictly μ @ 0K

(all states near E_c)

$$g_c'(E) = N_c\, \delta(E - E_c)$$



$$\int n(E)\,dE = \int g'(E) f(E)\,dE$$

$$= N_c \int \delta(E - E_c) \frac{1}{1 + e^{\left(\frac{E - E_F}{kT}\right)}}\,dE$$

$$= N_c \int \delta(E - E_c)\, e^{-\left(\frac{E - E_F}{kT}\right)}\,dE$$

$$= N_c\, e^{-\left(\frac{E_c - E_F}{kT}\right)}$$

As E_f goes up : More e^- in conduction band

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For holes in valence band:

prob of not occupying ↑

$$p = \int_{E_{V,bottom}}^{E_{V,top}} p(E)\,dE = \int_{E_{V,bottom}}^{E_{V,top}} g_v(E)\,(1-f(E))\,dE$$

$E_{V,top}$, $E_{V,bottom} \approx 12\,eV$

$$= \int_{-\infty}^{0} g_v(E)\,(1-f(E))\,dE$$

$$p = N_V e^{-\frac{(E_F - E_V)}{kT}} \quad ; \quad N_V = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

(no. of holes) ; N_V ↑ effective DOS for holes

$$\therefore N_V^{Si} \approx 3.1 \times 10^{19}/cm^3 \ (@\ 300K)$$

~~$m_p^* = 1.15\,m_0$~~
~~$m_n^* = 1.05\,m_0$~~
$m_p^* = 1.15\,m_0$
$m_n^* = 1.05\,m_0$

As E_F goes up: less holes in valence band

Intrinsic semiconductor:

$$n = p$$

$$\Rightarrow N_C e^{-\frac{(E_C - E_F)}{kT}} = N_V e^{-\frac{(E_F - E_V)}{kT}}$$

$$\Rightarrow e^{-\left(\frac{E_C - E_F}{kT}\right) + \left(\frac{E_F - E_V}{kT}\right)} = \frac{N_V}{N_C}$$

$$\frac{2E_F - E_C - E_V}{kT} = \ln\left(\frac{N_V}{N_C}\right)$$

$$\Rightarrow E_F = \frac{1}{2}\left[kT \ln\left(\frac{N_V}{N_C}\right) + E_C + E_V\right]$$

$$E_F = \frac{E_C + E_V}{2} + \frac{kT}{2}\ln\left(\frac{N_V}{N_C}\right)$$

At 0K; $E_F = \frac{E_c + E_v}{2}$

∴ At 0K; μ or fermi level lies at the middle of the band gap.

$n = p$;

$$np = N_c e^{-\frac{(E_c - E_F)}{kT}} N_v e^{-\frac{(E_F - E_v)}{kT}} = n_i^2$$

$$= N_c N_v e^{-\left(\frac{E_c - E_F + E_F - E_v}{kT}\right)} = N_c N_v e^{-\frac{E_g}{kT}}$$

$E_g \rightarrow$ Band Gap

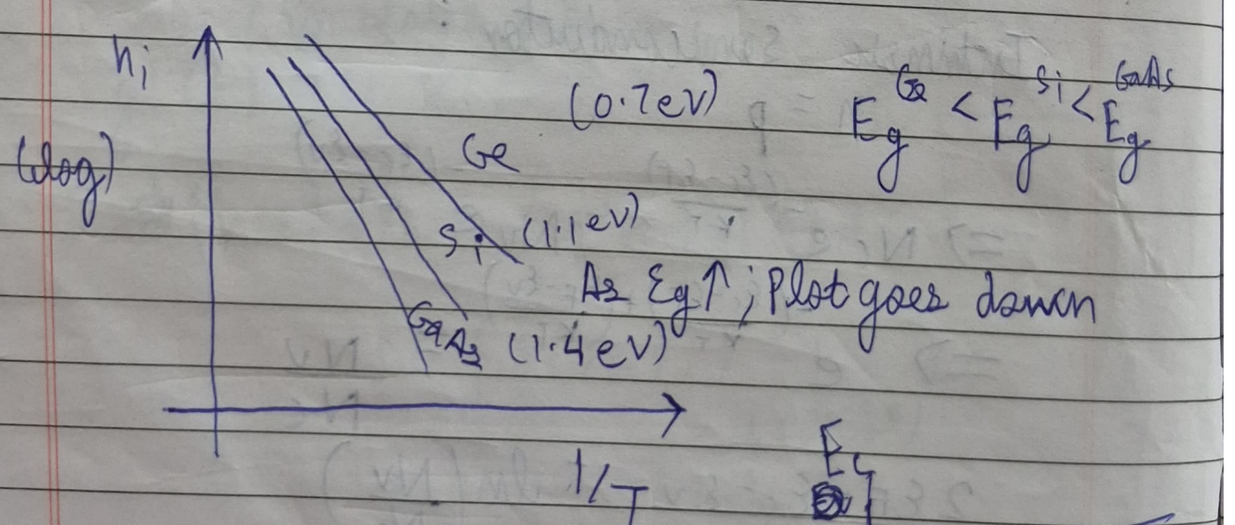
$$n = p = n_i = \sqrt{N_c N_v}\, e^{-\frac{E_g}{2kT}}$$

↳ Intrinsic carrier concentration

$E_g|_{Si}^{book} = 1.1\,eV$; $N_c = 2.86 \times 10^{19}$, $N_v = 3.1 \times 10^{19}$

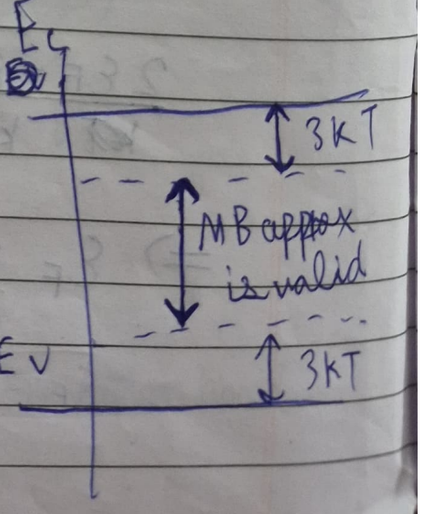
$kT = 26\,meV$

$n_i^{Si} \rightarrow \approx 1.07 \times 10^{10}/cm^3$



$n = N_c F_{1/2}(\eta)$; $P = N_v F_{1/2}(\eta)$

$$F_{1/2}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{\eta}}{1 + e^{(\eta - x)}}\, d\eta$$



Intrinsic Semiconductor : $\rho \downarrow$ as $T \uparrow$
For metals : $\rho \uparrow$ as $T \uparrow$ (resistivity)

Position of E_F :

$$E_F = \frac{E_c + E_v}{2} + \frac{KT}{2} \ln\left(\frac{N_v}{N_c}\right)$$

at $T = 0$: $E_F = \frac{E_c + E_v}{2} = E_{Fi}$

$$n = N e^{-\left(\frac{E_c - E_F}{KT}\right)} = N_c e^{-\left(\frac{E - E_F - E_{Fi} + E_{Fi}}{KT}\right)}$$

$$n_i = N_c e^{-\left(\frac{E_c - E_{Fi}}{KT}\right)} \Rightarrow n = N_c e^{-\left(\frac{E_c - E_{Fi}}{KT}\right)} e^{\left(\frac{E_F - E_F}{KT}\right)}$$

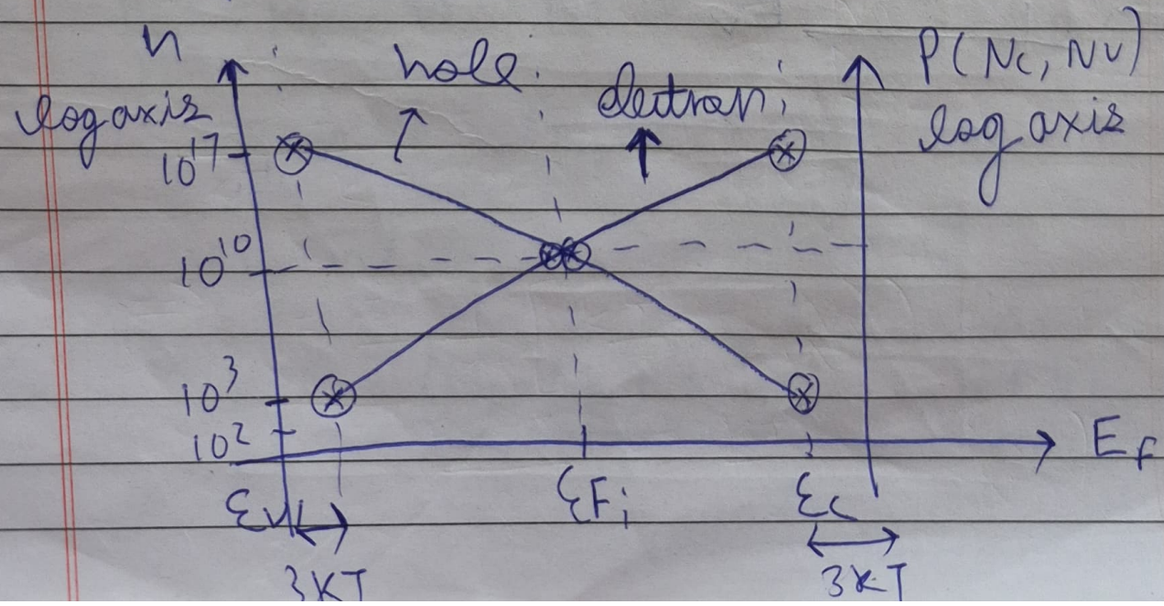
$$\boxed{n = n_i e^{\left(\frac{E_F - E_{Fi}}{KT}\right)}}$$

$$p = N_v e^{-\left(\frac{E_F - E_v}{KT}\right)} = N_v e^{-\left(\frac{E_F - E_v - E_{Fi} + E_{Fi}}{KT}\right)}$$

$$= N_v e^{\left(\frac{E_{Fi} - E_F}{KT}\right)} e^{-\left(\frac{E_{Fi} - E_v}{KT}\right)}$$

$$\boxed{p = n_i e^{\left(\frac{E_{Fi} - E_F}{KT}\right)}}$$

$$np = n_i^2$$

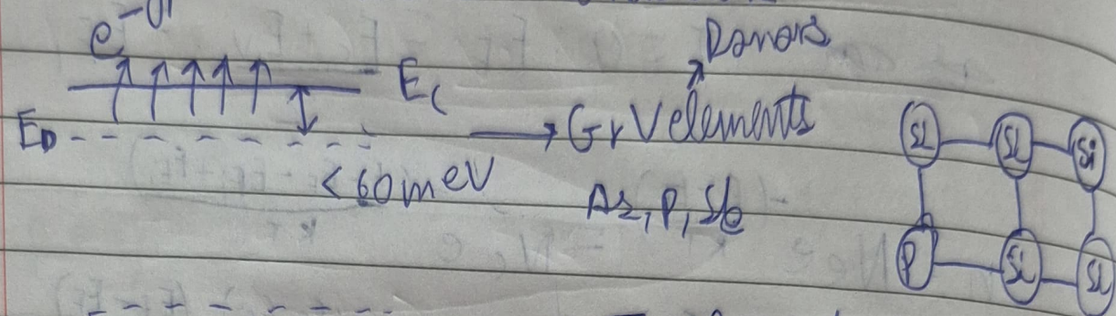


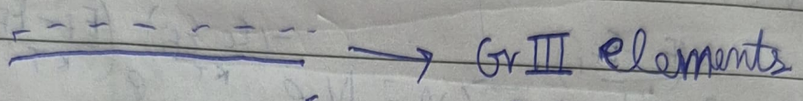
- Extrinsic Semiconductor

- Doping : Mechanism where selected foreign atoms are added in a controlled manner

- Thermal Diffusion, Ion-Implantation

n-type extrinsic semiconductors:

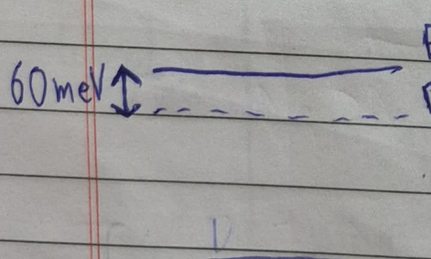




p-type extrinsic semiconductors

n-type : majority carrier electrons $n > p$

p-type : majority carrier holes $p > n$



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Two desirable properties of good Donor/Acceptor

1) Shallow Donor/Acceptor energy levels (< 60 meV)

2) High Solid ~~Solid~~ Solubility limit
$(10^{18} - 10^{19}) / cm^3$
$\sim 10^{20}$

Charge Neutralization

A homogeneous ~~solution~~ semiconductor with uniform doping

locally neutral

$P + N_D^+ = n + N_A^-$; $N_D \rightarrow$ Donor concentration ($/cm^3$)
(Full Ionization)

$N_A \rightarrow$ Acceptor Concentration ($/cm^3$)

$N_D^+ \rightarrow$ must fraction of N_D ionised

$N_A^- \leftrightarrow$ " " " N_A is ionised?

$P + N_D = n + N_A$ (Under full ionization)

$\frac{n_i^2}{n} + N_D = n + N_A$ $(N_D > N_A)$

$\Rightarrow n_i^2 + n(N_D - N_A) - n^2 = 0$

$n^2 - n(N_D - N_A) - n_i^2 = 0$

$$n = \frac{(N_D - N_A) \pm \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2}$$

$$n = \left(\frac{N_D - N_A}{2}\right) \pm \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2}$$

$$n = \left(\frac{N_D - N_A}{2}\right) + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \quad (+\text{ve root})$$

$$n = \left(\frac{N_D - N_A}{2}\right) + \left(\frac{N_D - N_A}{2}\right)\sqrt{1 + \frac{4n_i^2}{(N_D - N_A)^2}}$$

If $\frac{4n_i^2}{(N_D - N_A)^2} \ll 1$:

$$n = \left(\frac{N_D - N_A}{2}\right) + \left(\frac{N_D - N_A}{2}\right)\left(1 + \frac{1}{2}\frac{4n_i^2}{(N_D - N_A)^2}\right)$$

$\hookrightarrow \approx 1$

$$\Rightarrow n = N_D - N_A$$

$$p = \frac{n_i^2}{(N_D - N_A)}$$

If $N_A > N_D$ (p type):

$$p = N_A - N_D$$

$$n = \frac{n_i^2}{N_A - N_D}$$

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Charge Neutrality

$p - n + N_D^+ - N_A^- = 0$ (For uniformly doped semiconductor)

p – mobile at 300K; N_D^+, N_A^- – Immobile at 300K

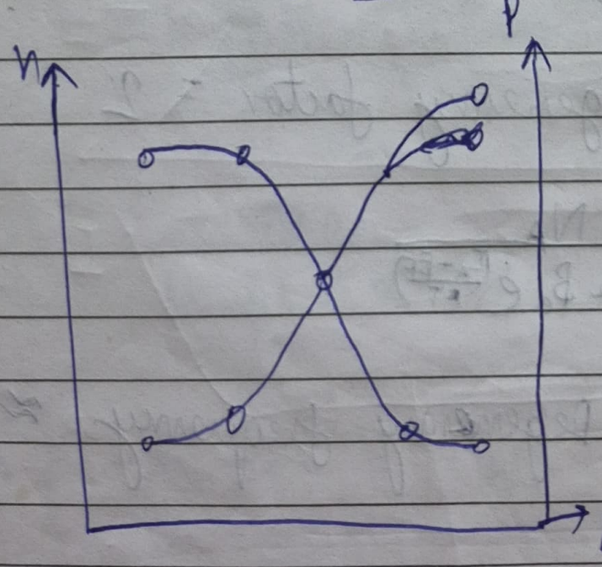
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2. $N_A = 10^{15}\ cm^{-3}$; $n_i = 10^{10}\ cm^{-3}$ at 300K

N_D	$n\ (cm^{-3})$	$p\ (cm^{-3})$	$E_F - E_{Fi}$ (meV)	
10^{13}	10^{15}	10^{15}	-299	($N_D < N_A$ so $p = N_A - N_D$)
10^{14}	1.1×10^{5}	9×10^{14}	-296	
10^{15}	$n_i = 10^{10}$	$n_i = 10^{10}$	0	
10^{16}	9×10^{15}	1.1×10^{4}	356	
10^{17}	10^{17}	10^{3}	419	

$$E_F - E_{Fi} = \frac{kT}{q} \ln\left(\frac{n}{n_i}\right)$$

$\frac{kT}{q} \approx 26mV$



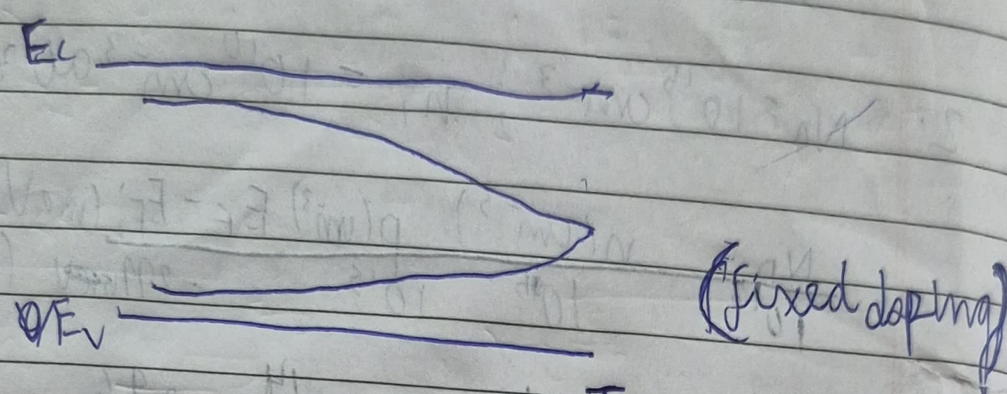
Effect of Temperature

$n = N_D$ at 300K

$$p = \frac{n_i^2}{N_D}$$

$n_i(T) \approx n$

$$n_i(T) = \sqrt{N_c N_v}\, e^{-\frac{E_g}{2kT}}$$



At low temperatures:

Partial Ionization (Carrier Freeze-out)

$$p - n + N_D^+ - N_A^- = 0$$

Full Ionization:
$N_D^+ = N_D$
$N_A^- = N_A$

Partial:

$$N_D^+ = \frac{N_D}{1 + B_d e^{\left(\frac{E_F - E_D}{kT}\right)}}$$

B_d = Donor Degeneracy factor = 2

$$N_A^- = \frac{N_A}{1 + B_a e^{\left(\frac{E_A - E_F}{kT}\right)}}$$

B_a = Acceptor Degeneracy frequency ≈ 4

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$$p - n + \frac{N_D}{1+\beta_D e^{\left(\frac{\varepsilon_F-\varepsilon_D}{kT}\right)}} - \frac{N_A}{1+\beta_A e^{\left(\frac{\varepsilon_A-\varepsilon_F}{kT}\right)}} = 0$$

$$N_V e^{-\left(\frac{\varepsilon_F-\varepsilon_V}{kT}\right)} - N_c e^{-\left(\frac{\varepsilon_c-\varepsilon_F}{kT}\right)} + \frac{N_D}{1+\beta_D e^{\left(\frac{\varepsilon_F-\varepsilon_D}{kT}\right)}} - \frac{N_A}{1+\beta_A e^{\left(\frac{\varepsilon_A-\varepsilon_F}{kT}\right)}} = 0$$

Assuming MB approx is valid:
Solve the eq to get position of fermi level

A-type Semiconductor:

$$p - n + N_D^+ - N_A^- = 0 \; ; N_A > 0$$

$n \approx N_D^+$ assuming p is very small

$$n = \frac{N_D}{1+\beta_d e^{\left(\frac{\varepsilon_F-\varepsilon_D}{kT}\right)}} \; ; = \frac{N_D}{1+\beta_d e^{\left(\frac{\varepsilon_F-\varepsilon_D+\varepsilon_c-\varepsilon_c}{kT}\right)}}$$

$$= \frac{N_D}{1+\beta_d e^{\left(\frac{\varepsilon_F-\varepsilon_c}{kT}\right)} e^{\left(\frac{\varepsilon_c-\varepsilon_D}{kT}\right)}}$$

$$n = \frac{N_D}{1+\frac{\beta_d}{N_c}\underbrace{\left(N_c e^{-\left(\frac{\varepsilon_c-\varepsilon_F}{kT}\right)}\right)} e^{\left(\frac{\varepsilon_c-\varepsilon_D}{kT}\right)}} \quad \underbrace{\qquad}_{\to n}$$

$$n = \frac{N_D}{1+\frac{\beta_d}{N_c} n e^{\left(\frac{\varepsilon_c-\varepsilon_D}{kT}\right)}} = \frac{N_D}{1+\frac{\beta n}{N_c} e^{\left(\frac{\varepsilon_D}{kT}\right)}}$$

$$n\left(1+\frac{\beta n}{N_c} e^{\left(\frac{\varepsilon_d}{kT}\right)}\right) = N_D$$

$$n + \frac{\beta n^2 e^{\frac{\varepsilon_d}{kT}}}{N_c} = N_D$$

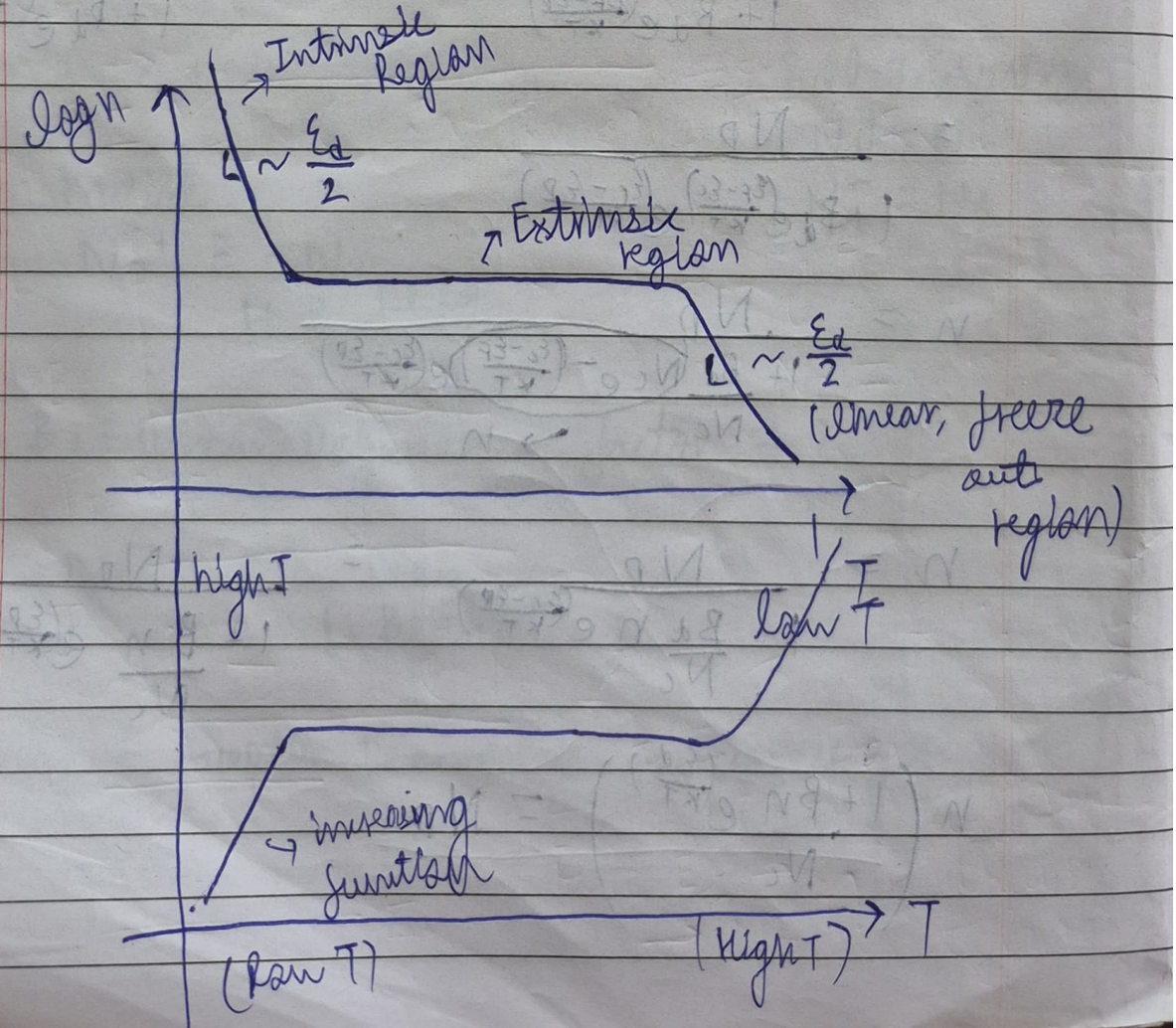
$$\Rightarrow n = \frac{-1 \pm \sqrt{1 + \frac{4 N_D \beta e^{\frac{\varepsilon_d}{kT}}}{N_c}}}{\frac{2\beta e^{\varepsilon_d/kT}}{N_c}}$$

$$\Rightarrow n = \frac{N_c}{2\beta_d} e^{\frac{-\varepsilon_d}{kT}} \left(\sqrt{1 + \frac{4\beta_d N_D e^{\frac{\varepsilon_d}{kT}}}{N_c}} - 1 \right)$$

$kT \ll \varepsilon_d$

$$n \approx \frac{N_c}{2\beta_d} e^{\frac{-\varepsilon_d}{kT}} \left(\sqrt{\frac{4\beta_d N_D}{N_c}} \right) e^{\frac{\varepsilon_d}{2kT}}$$

$$\boxed{n = \sqrt{\frac{N_c N_D}{\beta_d}} e^{\frac{-\varepsilon_d}{2kT}}}$$

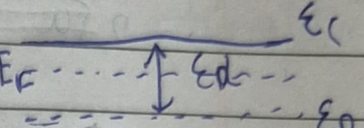


$$\sqrt{\frac{N_c N_D}{\beta_d}}\, e^{\frac{-\varepsilon_d}{2KT}} = n = N_c e^{-\frac{(\varepsilon_c - \varepsilon_F)}{KT}}$$

$$e^{\frac{\varepsilon_F}{KT}} = e^{\left(\frac{\varepsilon_c - \varepsilon_d}{2}\right)\left(\frac{1}{KT}\right)} \sqrt{\frac{N_D}{\beta_d N_c}}$$

$$\frac{\varepsilon_F}{KT} = \left(\frac{\varepsilon_c - \varepsilon_D}{2}\right)\frac{1}{KT} + \ln \sqrt{\frac{N_D}{\beta_d N_c}}$$

$$\varepsilon_F = \left(\varepsilon_c - \frac{\varepsilon_d}{2}\right) + KT \ln \sqrt{\frac{N_D}{\beta_d N_c}} \quad @0K$$



Tutorial - 6

1. $n(\varepsilon) = g_c(\varepsilon) f(\varepsilon)$; $E_c - E_F \geq 3KT$
$E_F - E_v \geq 3KT$

Degenerate Semiconductor:
Very heavily doped semiconductor:
Doping levels > N_c or N_v n^+, n^{++}
p^+ p^{++}

Degeneracy:
$H\psi_1 = E\psi_1$
$H\psi_2 = E\psi_2$ $\Rightarrow$ Degenerate Soln

$$g(\varepsilon) = a\sqrt{E - E_c} \; ; \; f(\varepsilon) = \frac{1}{1 + e^{\frac{\varepsilon - \varepsilon_F}{KT}}}$$

$$g(\varepsilon) f(\varepsilon) = a\sqrt{E - E_c}\, e^{-\frac{(\varepsilon - \varepsilon_F)}{KT}} \quad (MB\ Approx)$$

$$= a\sqrt{E - E_c}\, e^{-(\varepsilon - \varepsilon_c)} e^{-(\varepsilon_c - \varepsilon_F)}$$

$$= A'' \sqrt{E - E_c}\, e^{-\frac{(E - E_c)}{KT}} \; ; \; \frac{E - E_c}{KT} = u$$

$= A''\sqrt{kT}\sqrt{u}e^{-u} = B\sqrt{u}e^{-u}$

Differentiate: $\frac{1}{2\sqrt{u}}e^{-u} - e^{-u}\sqrt{u} = 0$

$\frac{1}{2\sqrt{u}} = \sqrt{u} \Rightarrow u = \frac{1}{2} \Rightarrow \frac{E-E_c}{kT} = \frac{1}{2}$

$\Rightarrow E = E_c + \frac{kT}{2}$; Holes: $E = E_v - \frac{kT}{2}$

3. (a) $n_i(T) = \frac{5}{100} \times N_D = \frac{5}{100} \times 10^{15}$ (neglecting band gap narrowing at high temp)

$\sqrt{N_c N_v}\, e^{\frac{-\varepsilon_g}{2kT}} = 5 \times 10^{13}$

$\sqrt{N_c(300)\times\left(\frac{T}{300}\right)^{3/2} \times N_v(300)\times\left(\frac{T}{300}\right)^{3/2}}\, e^{\frac{-\varepsilon_g}{2kT}} = 5\times10^{13}$

$\left(\frac{T}{300}\right)^{3/2}\sqrt{N_c(300K)N_v(300K)}\, e^{\frac{-\varepsilon_g}{2kT}} = 5\times10^{13}$

$\varepsilon_g = 1.1\,eV$

(b) $E_F - E_i = \ln\left(\frac{N_D}{N_i}\right)$; $n_i = 10^{10}$, $N_D = 10^{15}$

$E_F \approx E_i$

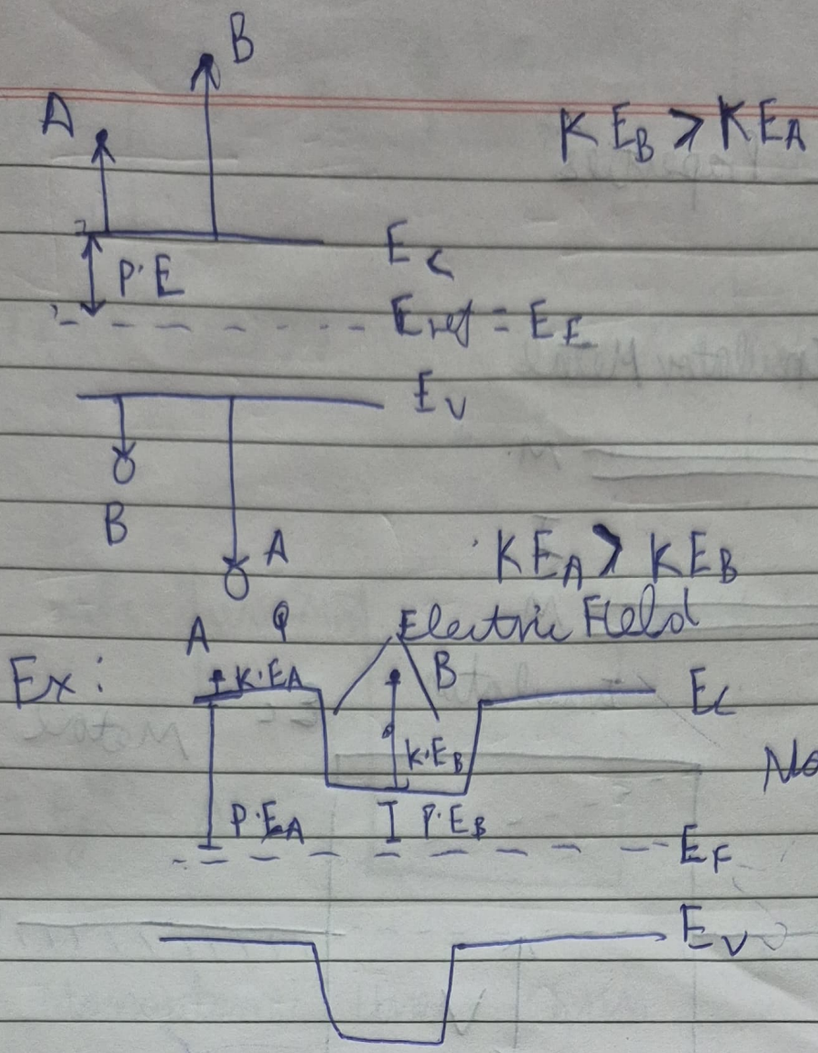
5. In equilibrium, there is only one fermi level for electron & holes

E_c — Flat E_c or E_v → no Electric field

E_F - - - - Flat E_F → equilibrium

E_v — → No current flowing.

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P $P \cdot E. = E_c - E_{ref} = U$ (eV)

Voltage (V) $= \frac{1}{-q}[E_c - E_{ref}]$

$E = -\frac{dV}{dx} = \frac{1}{q}\frac{d}{dx}[E_c - E_{ref}] = \frac{1}{q}\frac{dE_c}{dx}$

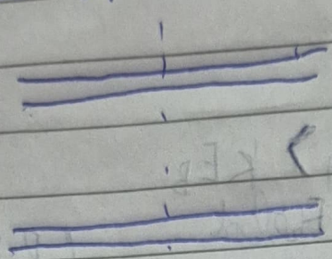
6. $E_c - E_F > 3kT$; $E_F - E_v > 3KT$ (or $\frac{E_g}{2} = 3KT$)

7. Take $E_g \leq 6KT$
↳ band gap

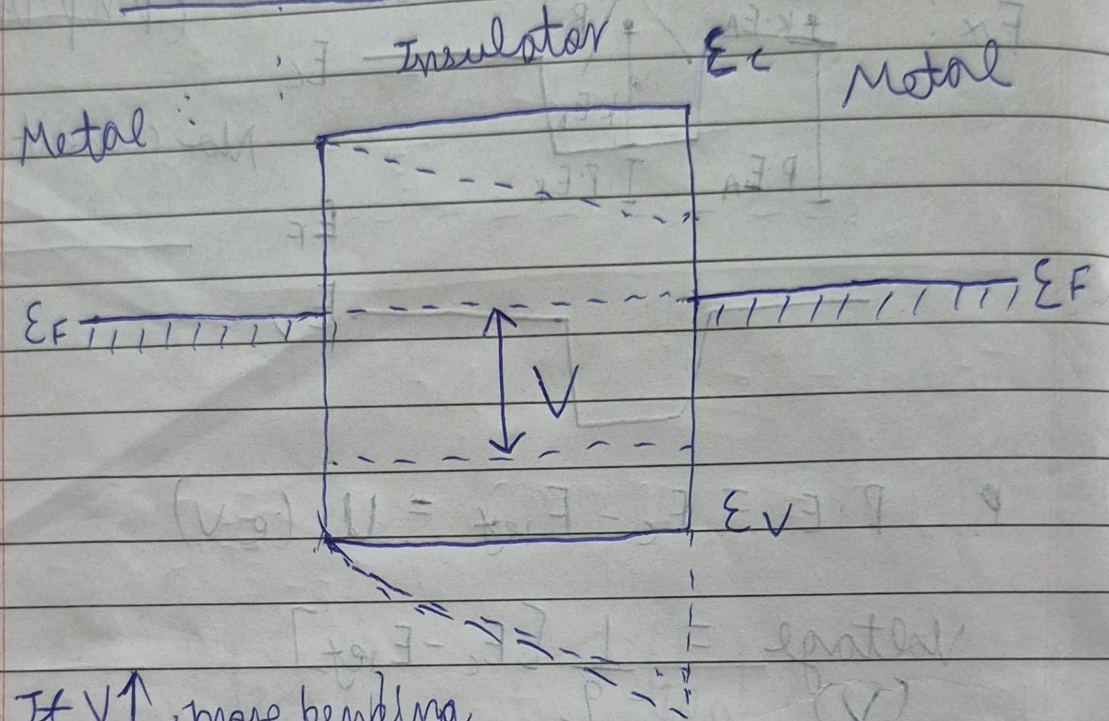
8. a) $\frac{1}{2}(E_D + E_c)$

Dielectric Properties

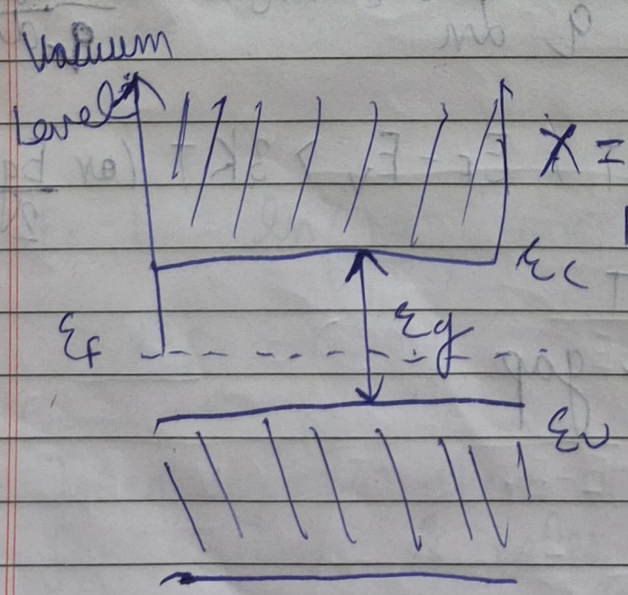
- Metal Insulator Metal



- Metal

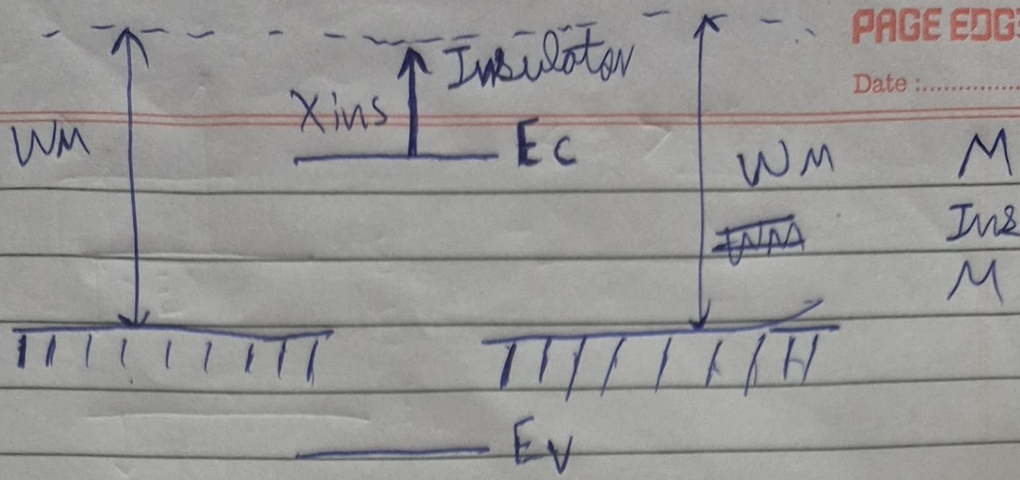


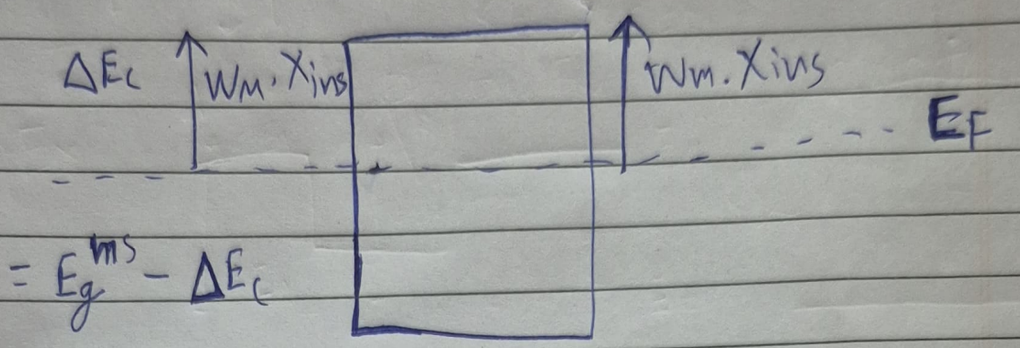
If V↑, more bending
⇒ More e^- can tunnel through barrier



E-x diagram
~4eV for Si
Electron affinity
= Vacuum level of E_c, Work function
= Vacuum level of E_F

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Monoatomic Linear Chain:

$$\omega = 2\sqrt{\frac{K}{m}}\,\sin\left(\frac{Ka}{2}\right)$$